

Udathu Manasa

✉ manasaudathu@gmail.com [in linkedin](#) [github](#)

EDUCATION

Gudlavalleru Engineering College

Bachelor of Technology in Computer Science & Engineering (CGPA: 9.4/10)

Jun. 2017 – Jul. 2021

Andhra Pradesh, India

- **Honors:** Awarded **Best Student of the Year (2021)** among 4,000+ students.

EXPERIENCE

ServiceNow

Software Engineer

Apr. 2024 – Present

Hyderabad, India

- Engineered and optimized backend pipelines for **Asset Assist (GenAI)**, integrating AI-powered search workflows, improving metadata quality and prompt relevance, implementing scheduled sampling jobs and error-handling mechanisms, and continuously refining production search accuracy across 5M+ asset records.
- Re-architected Oracle evaluation pipelines for 10M+ record datasets, eliminating expensive DB joins to **reduce execution time by 70%** while maintaining functional integrity.
- Resolved critical scalability bottlenecks in Software Asset Workspace, optimizing SQL execution plans and reducing response latency from **20s to 5s**.
- Improved **ML Normalization reliability** by implementing model-version fallback strategies and observability pipelines, preventing production failures.

Juspay

Software Development Engineer

Aug. 2022 – Feb. 2024

Bangalore, India

- Architected QR-code micro-apps and integrated web payment flows using **Android WebView** frameworks for high-concurrency merchant environments.
- Built and optimized merchant-facing dashboards using ReScript and React, enhancing UI responsiveness, stabilizing complex dropdown/filter workflows, resolving production UI issues, and implementing configurable merchant workflows, file upload/download capabilities, role-based action controls, and dynamic branding features across internal operational tools.
- Improved internal testing efficiency by **10x** through **PureScript** for dev-tooling refactors, resulting in a 50% reduction in QA overhead.
- Eliminated **80% of technical debt warnings** (1,700+ instances) by developing Python automation scripts, significantly enhancing codebase maintainability.
- Contributed to Cross-Border, Cross-Country, and Hybrid Pricing features by implementing configuration workflows and automated **Cucumber** validation suites.

Cognizant

Program Analyst

Jul. 2021 – Aug. 2022

Bangalore, India

- Rectified **50+ Veracode security violations** through static analysis remediation and secure coding practices across ASP.NET Core applications.
- Resolved Section 508 accessibility compliance issues using **WCAG 2.1** standards and ARIA attributes to ensure inclusive user experiences.
- Optimized database performance by analyzing indexing strategies and query execution across varied load conditions.

PROJECTS

Debug-Env: Distributed Systems RL Sandbox | Python, FastAPI, Docker, Reinforcement Learning

May. 2026

- Developed an OpenEnv-compatible simulator for distributed systems to evaluate LLM reasoning in **Root Cause Analysis (RCA)**; secured a spot in the **Top 2.5%** at the Meta x Scaler OpenEnv Hackathon.
- Engineered a custom **Feedback & Reward Loop** to optimize the agent's troubleshooting path by evaluating diagnostic steps against system state recovery.
- Containerized complex failure scenarios using **Docker**, enabling reproducible sandboxed environments for testing agent resilience against network latency and service outages.

Achievements

Global & National Rankings: **Top 2.5%** (Top 800) in Meta x Scaler OpenEnv Hackathon among 32,000+ teams; Secured **Rank 99** among 1.5L+ participants in Codekaze 2024 and **National Rank 5** in 2022.

Competitive Programming: Google Hash Code **Rank 3787**; Google Code Jam I/O for Women **Rank 2908**; Secured **AIR 6000** in GATE CSE.

SKILLS

Languages : Python, JavaScript, Java, ReScript, C, SQL, HTML/CSS

Frameworks & Tools : React, FastAPI, ASP.NET Core, Docker, Git, TailwindCSS

Core Concepts : Backend Systems, Distributed Systems, System Design (HLD/LLD), REST APIs, Performance Optimization, Scalability, Workflow Automation, Database Optimization, Observability, AI-Powered Workflows, RCA